Course Title	Feedback Control Systems	**Course Code**	EE3004
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Research Literature (WK8)	**Credit Hrs.**	3+1
Knowledge Area	Control Engineering (KA03)	**Grading Scheme**	Relative
HEC Knowledge Area	Electrical Engineering Core (Breadth)	**Applicable From**	Spring 2023
SDG	4 | Quality Education	**PBL**	1
Pre-requisite(s)	EL2008, EE2008	**CEP**	1

Course Objective	Students should understand the relationship between transfer function and stability of the system using classical tools like Routh-Hurwitz criterion, root locus, and Bode plots. They should also understand time-domain system analysis using state-space representation.

No.	Assigned Program Learning Outcome (PLO)
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Develop the mathematical model of a physical system	Q1, M1	C5	2
2	Analyze the transient and steady state response of a feedback system.	Q2, M2, F	C4	2
3	Analyze the stability of a system.	A1, M2, F	C4	2
4	Analyze a given system in frequency domain.	Q3, F	C4	2
5	Design a closed-loop feedback controller	A2, F	C5	3

Text Books	**Title**	Control Systems Engineering 6th Edition
	Author	Norman S. Nise
	Publisher	John Wiley & Sons Inc.
Reference Books	**Title**	Modern Control Engineering 5thEdition
	Author	Katsuhiko Ogata
	Publisher	Pearson Education

Week	Course Contents/Topics	Chapter*	CLO*
1	Basic Control System Concepts , System Configurations (Open loop systems, Closed-loop/feedback control systems), Automatic Control Systems, Laplace transform review in detail, Transfer Function and Impulse Response Function and Mathematical Modeling of Control Systems	1	1
2	Mathematical Modeling of Physical Systems	2	1
3	State equations for physical systems, applying the state-space representation	3	1
4	State equations for physical systems (Cont.)	3	1
5	Transient Response: Poles, zeros and system response First Order Systems	4	2
6	Second order systems, block diagram reduction	4, 5	2
7	Stability of systems	6	3
8	Steady state errors analysis in feedback control systems	7	2
9	Root Locus techniques Analysis of Systems using Root Locus	8	4
10	Transient Response design via gain adjustment root locus	8	4
11	Control systems design by the root locus method, preliminary design consideration, improving steady state error via cascade compensation, Lag compensation, PI controller	9	5
12	Improving transient response via cascade compensation, lead compensation, PD controller	9	5
13	Lag Lead compensation, PID controllers, physical realization of compensation	9	5
14	System analysis using bode plots, phase margin and gain margin	10	4
15	Compensator design using bode plot	10	5
16	Polar plots, Nyquist stability criterion, stability analysis	10	4

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes + Assignments	20.0%
Midterm (I, II)	30.0%
Final Exam	50.0%